

Introduction

Measles is a highly transmissible viral disease that can cause serious complications and death [1]. Although the UK achieved measles elimination in 2016, cases and outbreaks have increased in recent years alongside declining MMR vaccination coverage. Substantial inequalities exist in MMR coverage and measles infection. In March 2026, MMR coverage among Black African and Black Caribbean children in Birmingham and Solihull (BSol) was 74.2% and 64.0%, respectively, compared with 92.9% among White British children [2]. During the 2023/24 Birmingham outbreak, the measles infection rate among Black African residents was eight times that of the White British population [3].

A rapid literature review identified a complex range of systemic, social and psychological factors influencing childhood vaccination, informing the stakeholder workshop and model development.

Aim: To investigate how a stakeholder-informed system dynamics model can support the development of interventions to increase MMR vaccination coverage and reduce outbreak burden.

Methods

Stakeholders from BSol ICB, local authority public health teams, and UKHSA participated in a workshop to identify barriers and facilitators to childhood MMR vaccination and prioritise potential interventions to inform future model development.

The model builds on the model of Ackere and Schulz to incorporate five population states (susceptible, infected, isolated, hospitalised and recovered/vaccinated) across three age groups: babies (<1 year), children (1–4 years) and “adults” (5+ years). The model includes international migration and age-specific infection and hospitalisation rates [4]. For simplicity, individuals were considered completely immune to measles infection following a single dose of MMR.

The main model included three free parameters: the delay to measles isolation p_D , the proportion of infected individuals isolated each day p_P , and the hospitalisation rate for infected adults p_H . These parameters were calibrated against weekly measles cases and hospital admissions from the 2023/24 BSol outbreak using 8,000 simulations and a combined error function.

The calibrated model was then used to explore how varying levels of annual MMR vaccination and different outbreak management scenarios affect the estimated number of measles infections and hospital admissions.

System dynamics modelling was performed using Stella Architect and data analysis and visualisation was performed in R. All code and system dynamics models are publicly available on GitHub.

Results

The calibrated model reproduced the observed epidemic curve and hospital admissions, with modelled totals reaching 92% and 98% of observed values, respectively. The age distributions of infections and hospital admissions were also broadly consistent.

Outbreak Management

It was estimated that reducing the delay to isolation substantially reduced outbreak burden, whereas increasing the isolation rate had a smaller effect (Figure 1).

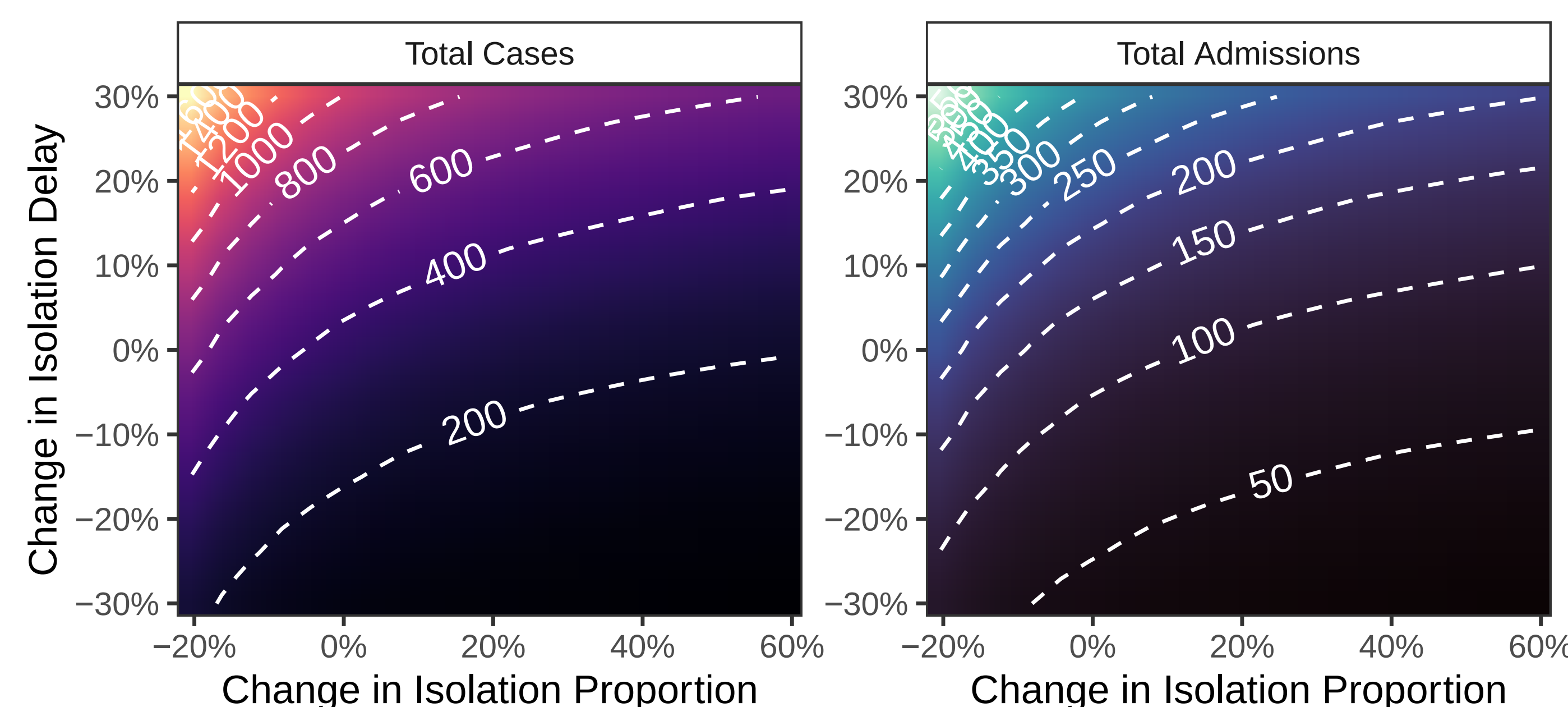


Figure 1. Estimated measles cases (left) and hospital admissions (right) under different isolation delays and isolation proportions, relative to the 2023/24 outbreak

Outbreak preparation

Changing the annual rate of childhood MMR vaccination produced a non-linear relationship with outbreak burden (Figure 2). Reductions in vaccination produced disproportionately larger increases in infections and hospitalisations, particularly when sustained over longer periods before an outbreak.

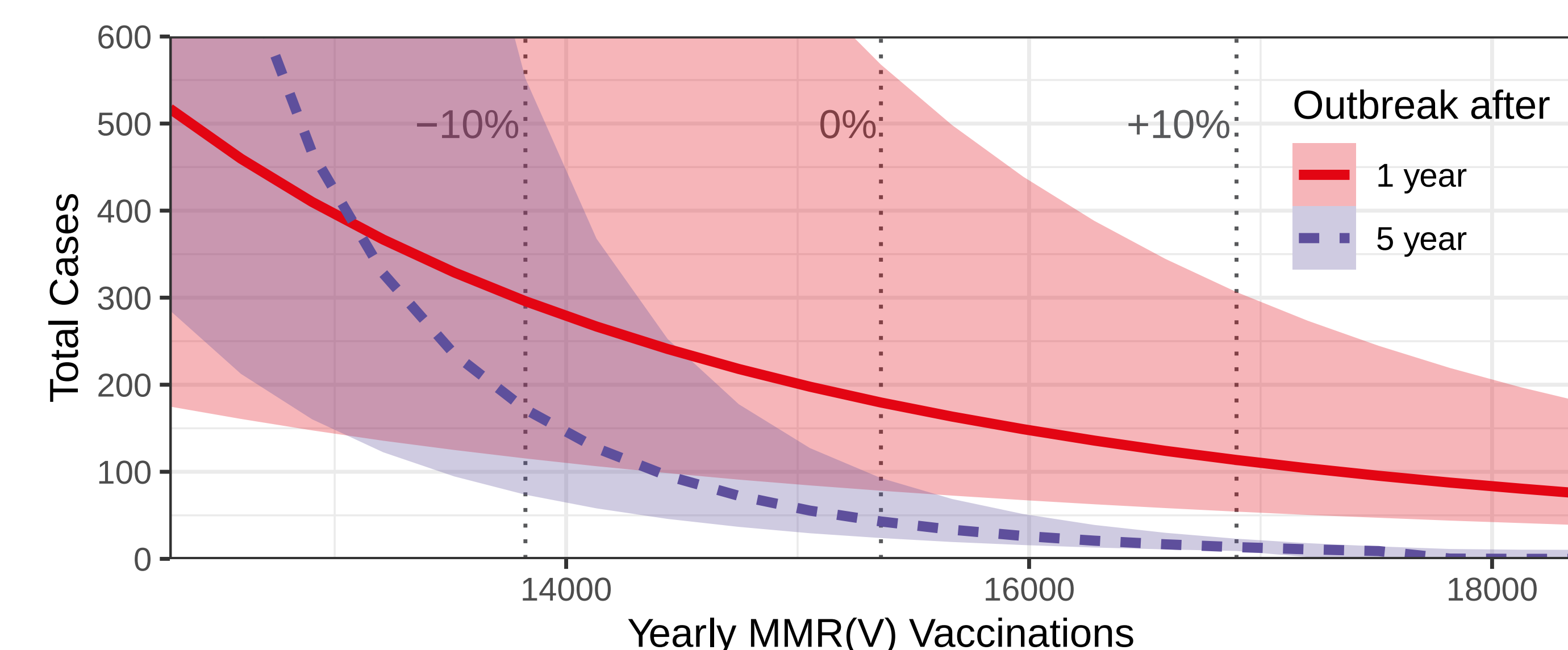


Figure 2. Estimated measles cases under different annual MMR vaccination rates, following one or five years of reduced vaccination.

Limitations

The model was calibrated against the 2023/24 outbreak, so external validity remains to be established. The model assumes homogeneous population mixing which does not capture geographic/ethnic heterogeneity in transmission. Additionally, current vaccination implementation may overestimate future coverage.

Summary & Conclusions

This study presents preliminary findings from an ongoing project that will contribute to future work packages for the BSol Measles Elimination Group. The calibrated model was able to reproduce the epidemic curve and hospital admissions used for calibration and demonstrated the potential impact of outbreak preparation and management on future burden. Delays in isolation were particularly important, and sustained reductions in MMR coverage could disproportionately increase future outbreak burden.

Future versions will incorporate ethnicity, stakeholder-identified behavioural factors and potential interventions to investigate their impact on vaccination inequalities and outbreak burden.

References

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- [4] Ann van Ackere and Peter J Schulz. Explaining vaccination decisions: A system dynamics model of the interaction between epidemiological and behavioural factors. *Socio-Economic Planning Sciences*, 71:100750, 2020.